Fiscal Year:	FY 2018	**Task Last Updated:**	FY 01/25/2022
PI Name:	Sielecki-Dzurdz, Thais Ph.D.		
Project Title:	LGM2605 as a Mitigator of Space Radiation-Induced Vascular Damage		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19104-5501	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	06/11/2018	**End Date:**	06/10/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC18C0089		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	LignaMed, LLC is developing LGM2605, an oral small molecule for use as a radiation mitigating agent. Here we aim to evaluate LGM2605 as a mitigator of space-radiation induced damage. NASA missions to Mars will expose astronauts to solar/galactic cosmic mixed radiation including low dose g and proton radiation, a source of harmful short and long-term health effects. Damage to the vascular network under mixed radiation types is not understood. Findings from our NASA-funded Phase I studies provided novel evidence that LGM2605 is an effective mitigator of radiation toxicity in cells exposed to mixed-field space-relevant radiation (high Linear Energy Transfer (LET) protons and gamma rays). In this application, LignaMed in collaboration with the researchers at the University of Pennsylvania will extend these initial studies to evaluate LGM2605 in an in vivo model for protection from radiation-induced i) carcinogenesis in lung, liver, and all major organs (Task 1) and accelerated lethality as a secondary endpoint and ii) tissue degeneration (Task 2) by evaluating long term lung deterioration and long-term damage mixed gender adult mice. We hypothesize that		

	mixed space radiation increases cancer risk and induces chronic, pro-inflammatory changes in tissues leading to accelerated degeneration of the cardiovascular and pulmonary system. We propose that LGM2605 will mitigate space radiation-induced carcinogenesis and tissue degeneration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	POTENTIAL NON-NASA COMMERCIAL APPLICATIONS: Lignamed LLC is a biopharmaceutical company developing LGM-2605 as adjunct therapy to reduce side effects and improve cure rates of radiation treatment of chest cancers. Chest cancers are a deadly and costly disease. They include breast cancer, lung cancer, sarcomas, lymphomas, and esophageal cancer. According to the American Cancer Society, more than 500,000 new chest cancer cases will be diagnosed in the United States in 2014 and they project the number to increase in the years ahead. About 50 to 60 percent of cancer patients are treated with radiation at some time during their disease. Combinations of surgery, chemotherapy, and radiation treatments are the standard for modern cancer therapy. Success is often determined by the ability of patients to tolerate the most aggressive regimen. The ability to deliver effective radiation therapy is limited by toxic side effects to healthy normal lung tissues. These side effects often cause breaks in treatment or dose-limiting toxicity after treatment, and, therefore, limit the amount of radiation that can be delivered to the tumor. No current therapies are effective to protect healthy normal lung tissue from the damaging effects of radiotherapy. A significant unmet need exists for a safe radioprotection agent that will ameliorate radiation side effects to normal tissue without "protecting" the tumor.
Task Progress:	New project for FY2018. Reporting not required for this SBIR Phase II project. (Ed. note: added to Task Book on 01/20/2022)
Bibliography Type:	Description: (Last Updated:)